Image Analysis and Computer Vision

Image Analysis and Computer Vision - teachers

Teacher: Prof. Vincenzo Caglioti vincenzo.caglioti@polimi.it

Exercise classes plus some theory will be given by:
Prof. Giacomo Boracchi giacomo.boracchi@polimi.it
Prof. Luca Magri luca.magri@polimi.it
Prof. Federica Arrigoni federica.arrigoni@polimi.it

Image Analysis and Computer Vision – weekly schedule

Tuesday	h 12.15 - 14.15	room 2.0.1
Wednesday	h 12.15 – 14.15	room 3.0.2

NO streaming
SOME recording will be available

FRAMEWORK

Observation of a static or dynamic scene in the 3D world

through

one or several 2D images

IACV main object

IMAGE ANALYSIS:

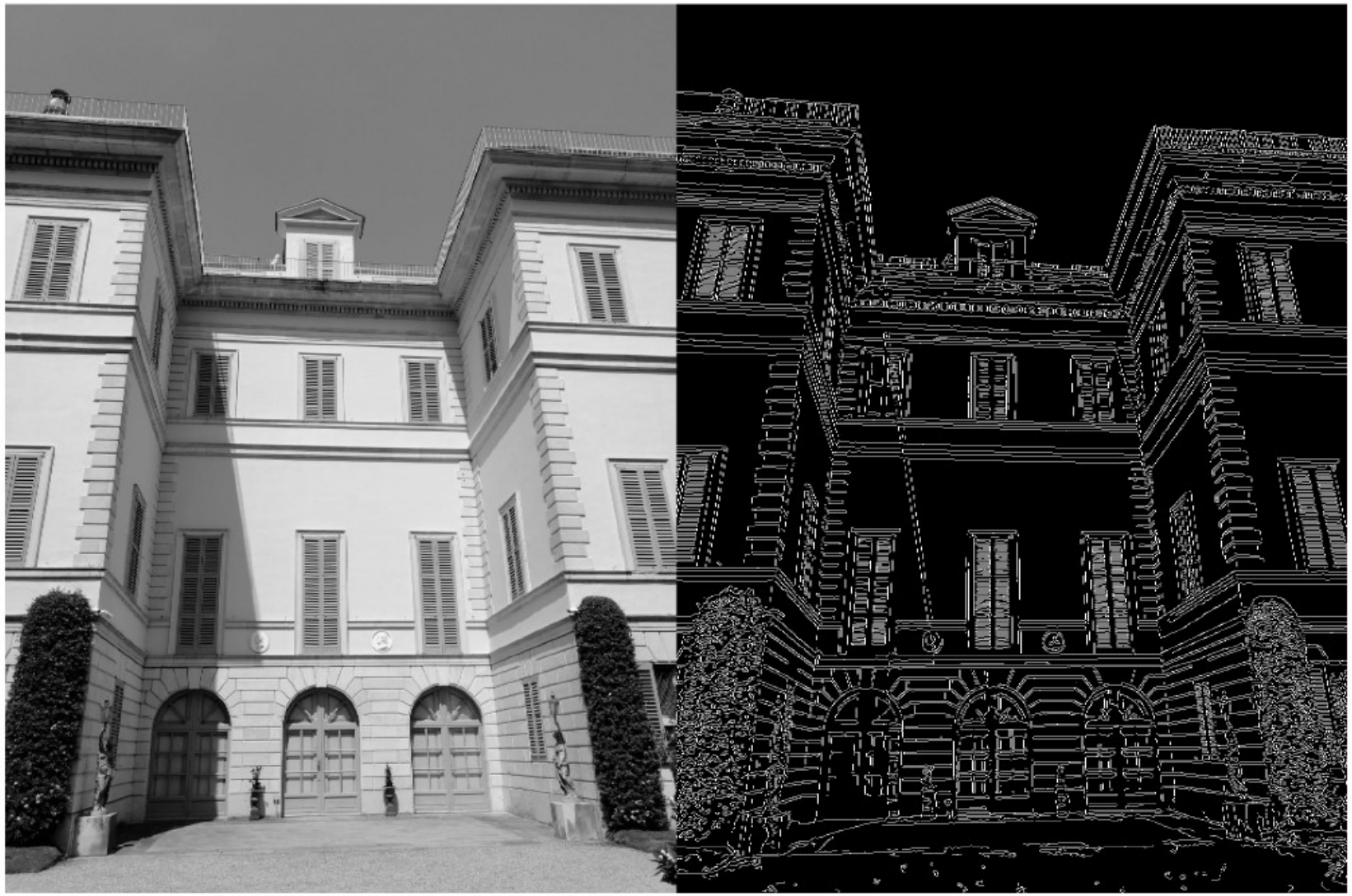
- extract image features (edges, corners, regions)
- match features among different images
- track features along an image sequence (video)

COMPUTER VISION:

compute a model of the observed scene

- 2D shape reconstruction
- 3D shape reconstruction
- Object recognition
- Object localization
- Action recognition

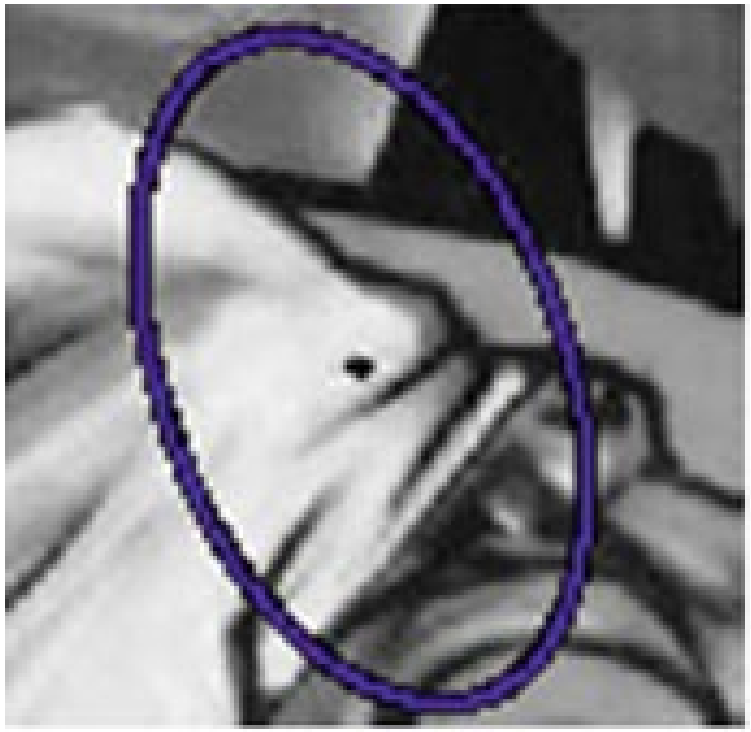
feature extraction: edges

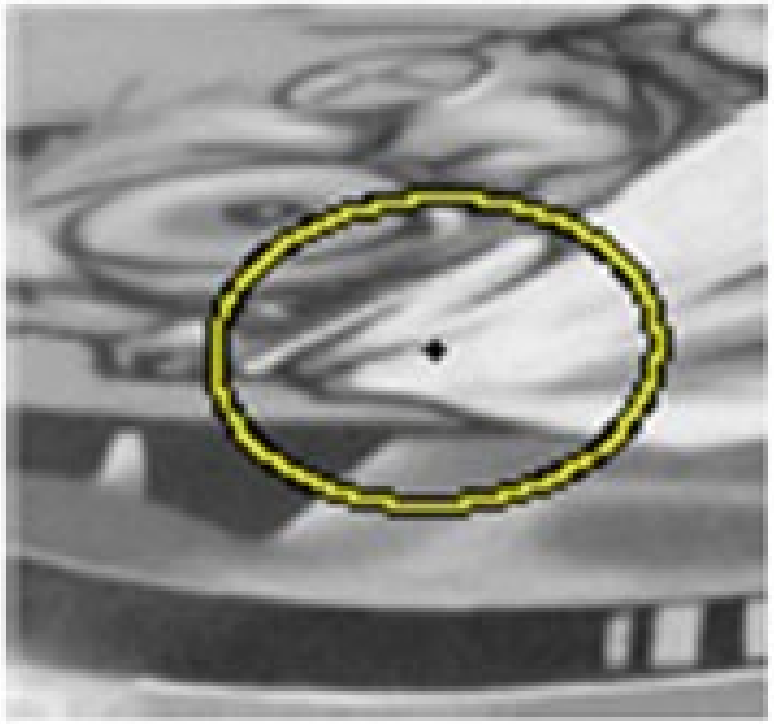


feature extraction: corners

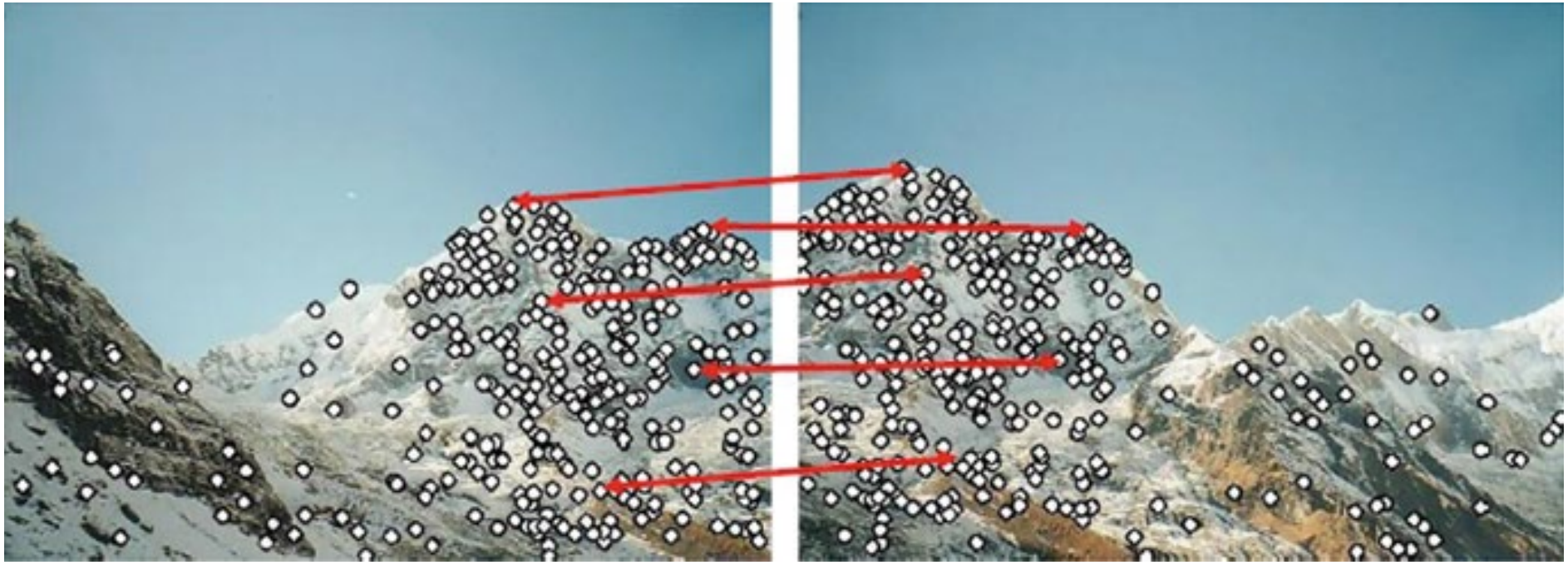


feature extraction: regions

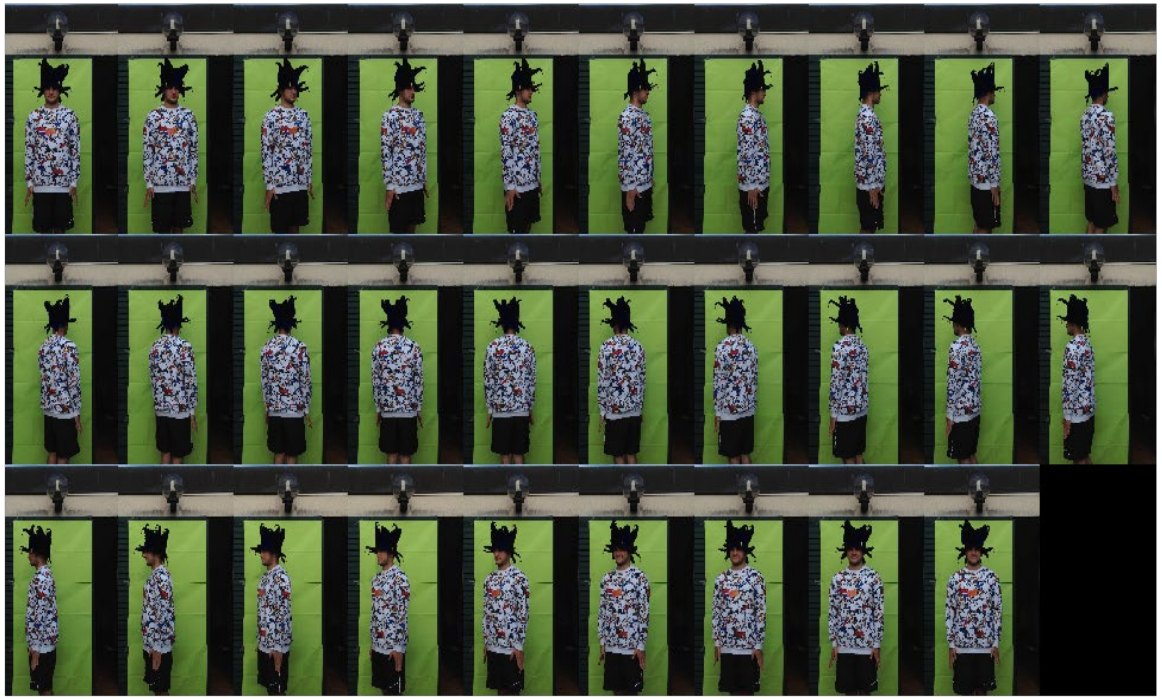


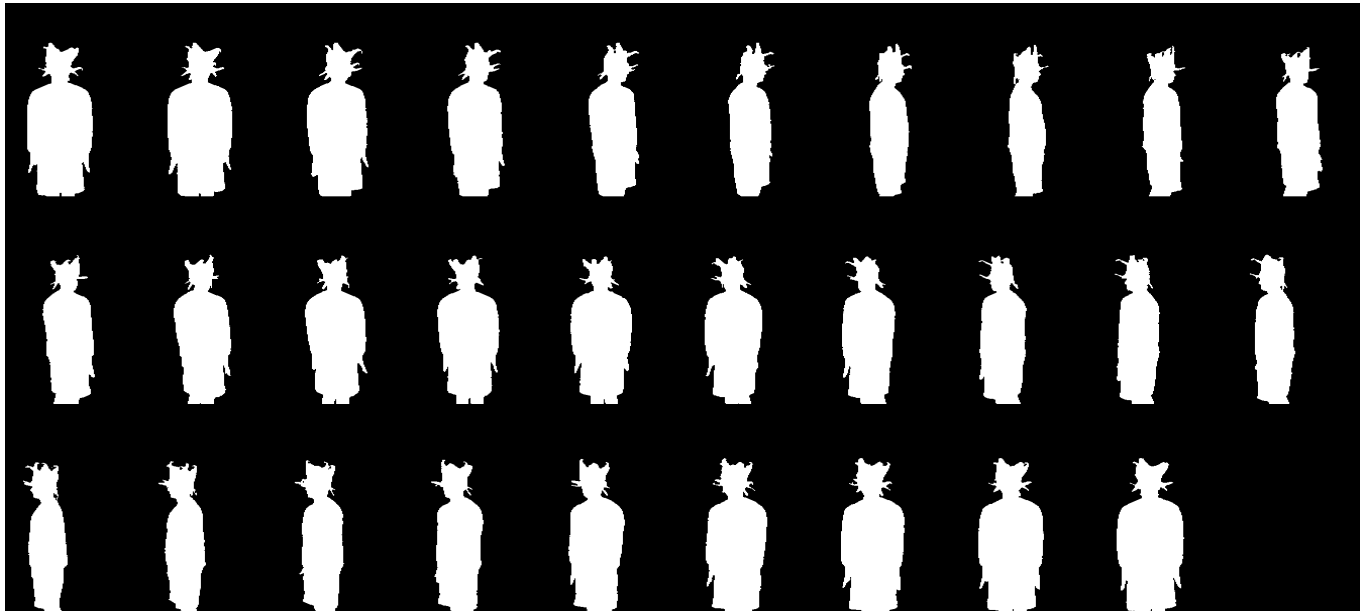


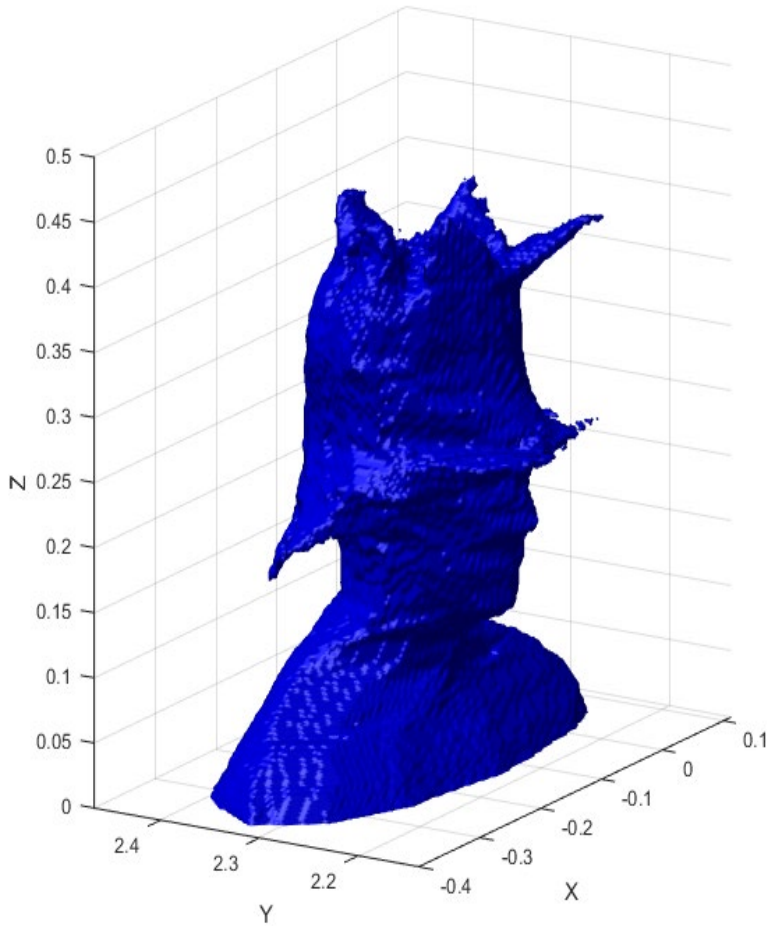
feature matching

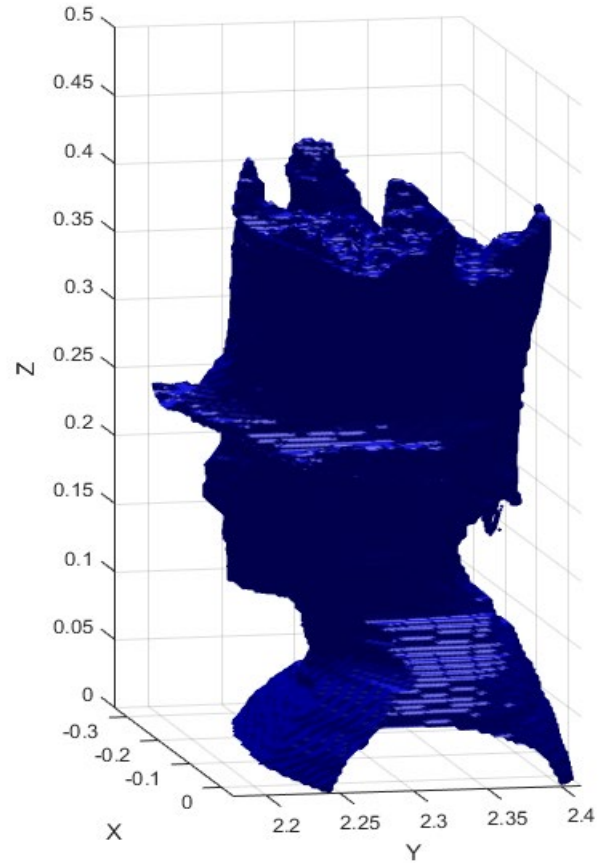


3D shape reconstruction






0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
Z
2.4
2.3
2.2
Y
-0.4
-0.3
-0.2
-0.1
0
0.1
X


0.5
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
Z
-0.3
-0.2
-0.1
0
X
2.2
2.25
2.3
2.35
2.4
Y

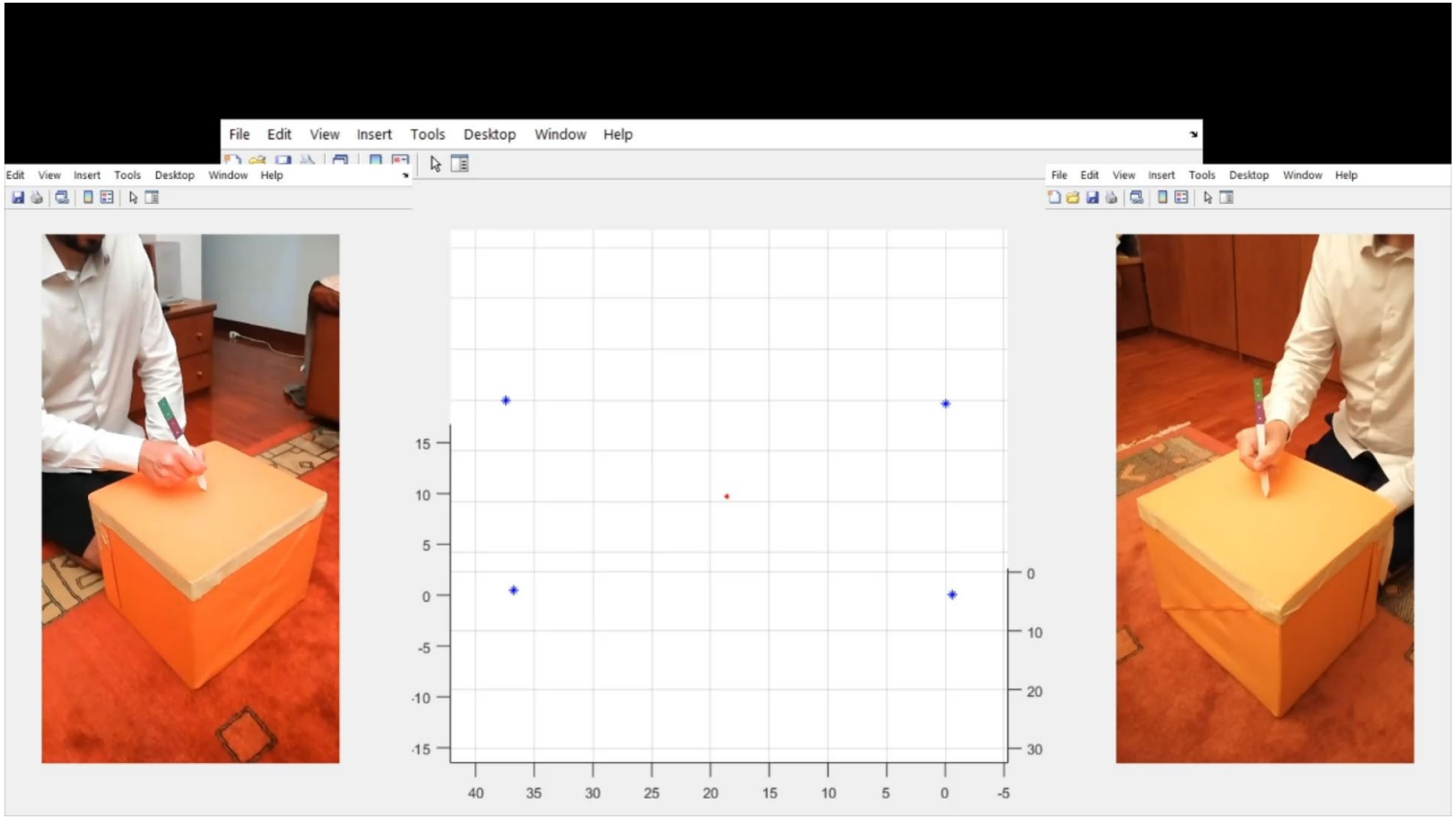


Microsoft

Object recognition



Object localization



Action recognition



DIFFICULTIES

- Noise, degradation, blur, and ILL-POSEDness

→feature extraction and feature matching difficult

- Projection -> information loss (dimensionality reduction from 3D to 2D)

→3D reconstruction is difficult

→

we need to study:

- IMAGE SIGNAL ANALYSIS
- PROJECTION GEOMETRY

TOPICS

- Introduction
- Optical Sensors (camera)
- 2D Projective Geometry (image plane, 2D planar scenes)
- 3D Projective Geometry (3D space scenes)
- Camera Geometry (3D → 2D projection) and Calibration
- Digital image Filters and Image Morphology
- Edge and Corner Detection
- Image Feature Detection and Description and Matching
- Model Fitting to Noisy Data
- Multi-view Geometry and 3D Shape Reconstruction
- …..

textbooks

- Hartley & Zisserman «Multi-view Geometry in Computer Vision»
- Forsyth and Ponce «Computer Vision: A Modern Approach»
- Course Slides on the WeBeep Platform

TOPICS

- Introduction
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- 2D Projective Geometry (image plane, planar scenes)
- 3D Projective Geometry (3D scenes)
- Camera Geometry (3D → 2D projection) and Calibration
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- Edge and Corner Detection
- Image Feature Detection and Description and Matching
- Model Fitting to Noisy Data
- Multi-view Geometry and 3D Shape Reconstruction
- ….. MOST DIFFICULT TOPIC: RULES TO PASS THE EXAM

FIRST RULE

ENROLL to the exam call,

in which you intend to take the exam

Usual deadline:

three working days before the call date

TWO DIFFERENT WAYS to take the exam

Standard exam → MAX grade: 30 cum laude

individual Homework & FULL Project

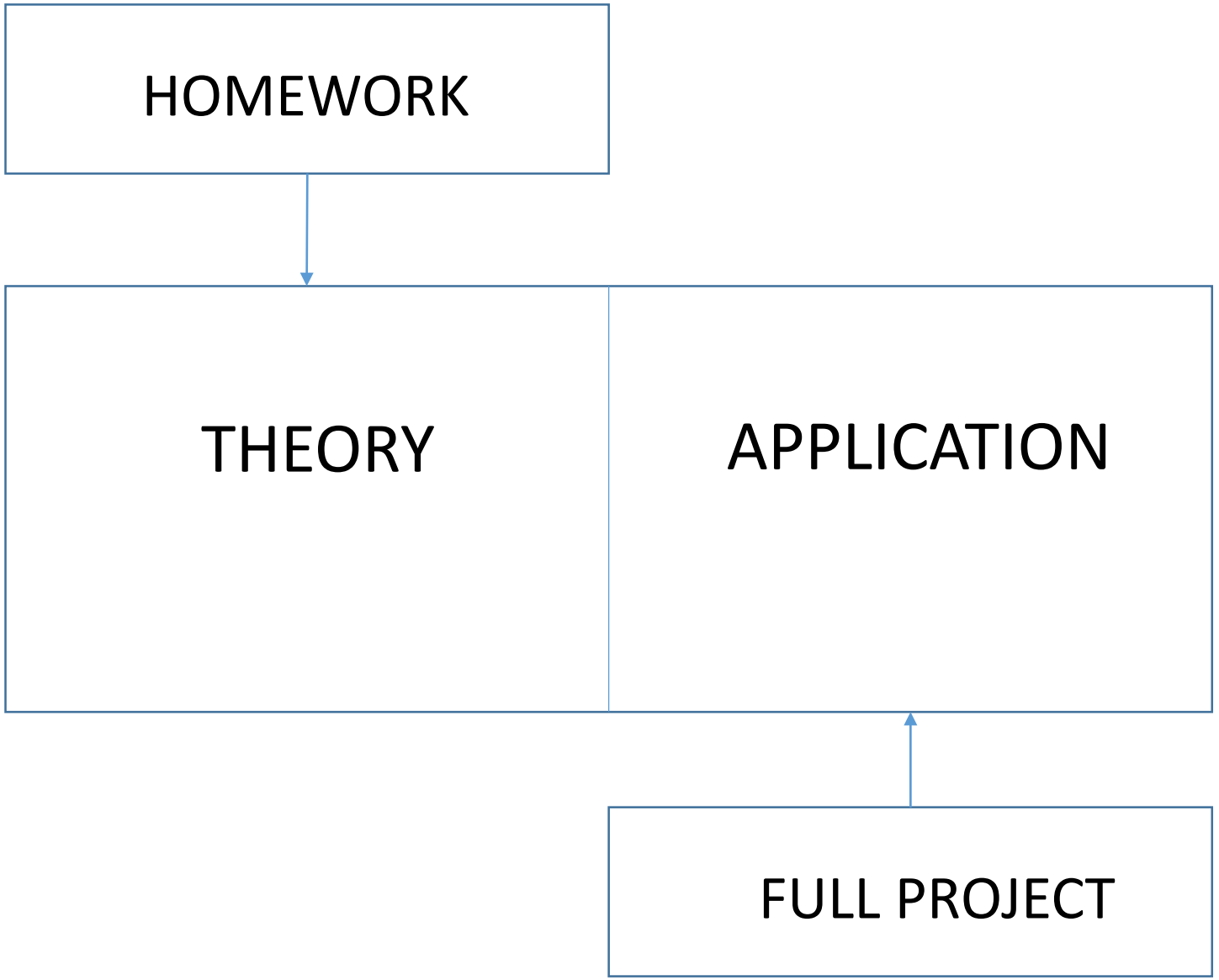
<u>ON REQUEST</u>: Non-standard exam → MAX grade: < 30

- Homework & SHORT Project → 26
- Project only (no Homework) + written proof → 28
- Homework only (no Project) + written proof → 27
- No Homework & no Project + written proof → 25

STANDARD EXAM: 2 parts → up to 30 cum laude

THEORY	APPLICATION

STANDARD EXAM: 2 parts → up to 30 cum laude



STANDARD EXAM → up to 30 cum laude

- **individual HOMEWORK** on Theory → max grade **10**/30:

  a long, **difficult** exercise (usually) about 2D reconstruction, camera calibration and localization + a little bit of image feature extraction

  EVALUATION BASED on (i) SUBMISSION and (ii) REPORT

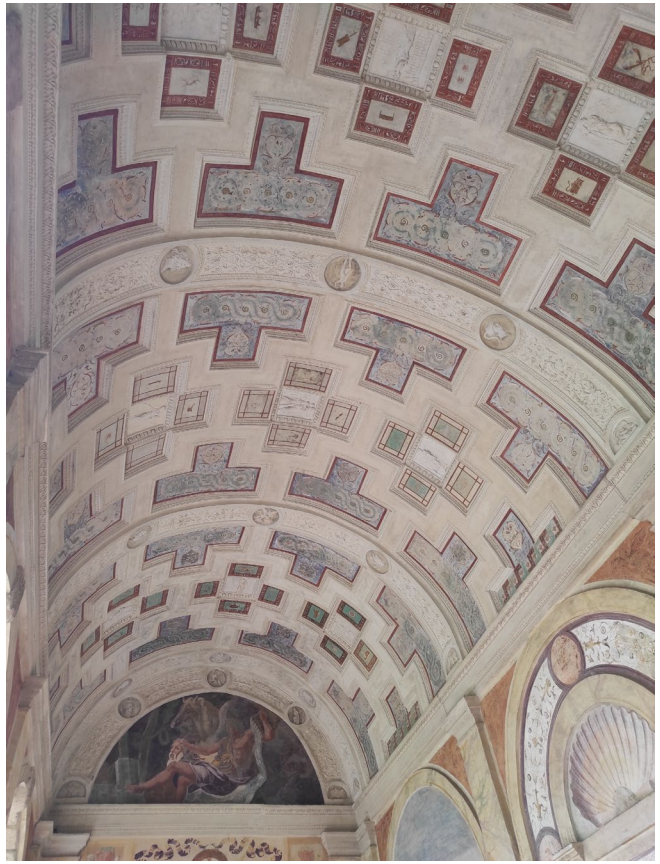
- either **individual or team FULL PROJECT** → max grade **20**/30:

  an individual or team work involving design, implementation and experimental activity

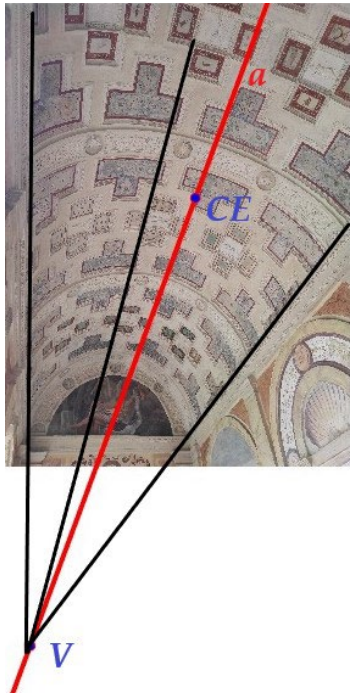
  EVALUATION BASED on

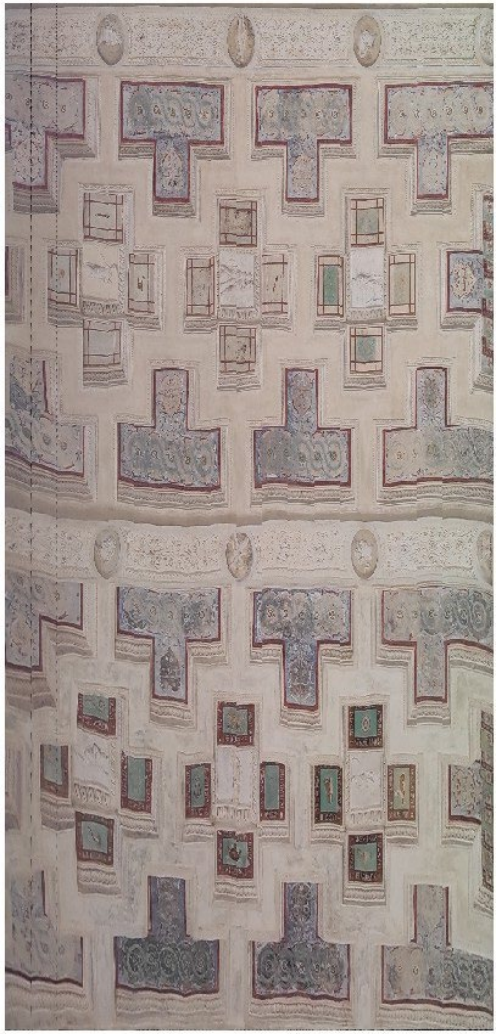
  (i) SUBMISSION, (ii) REPORT and (iii) PRESENTATION/DISCUSSION

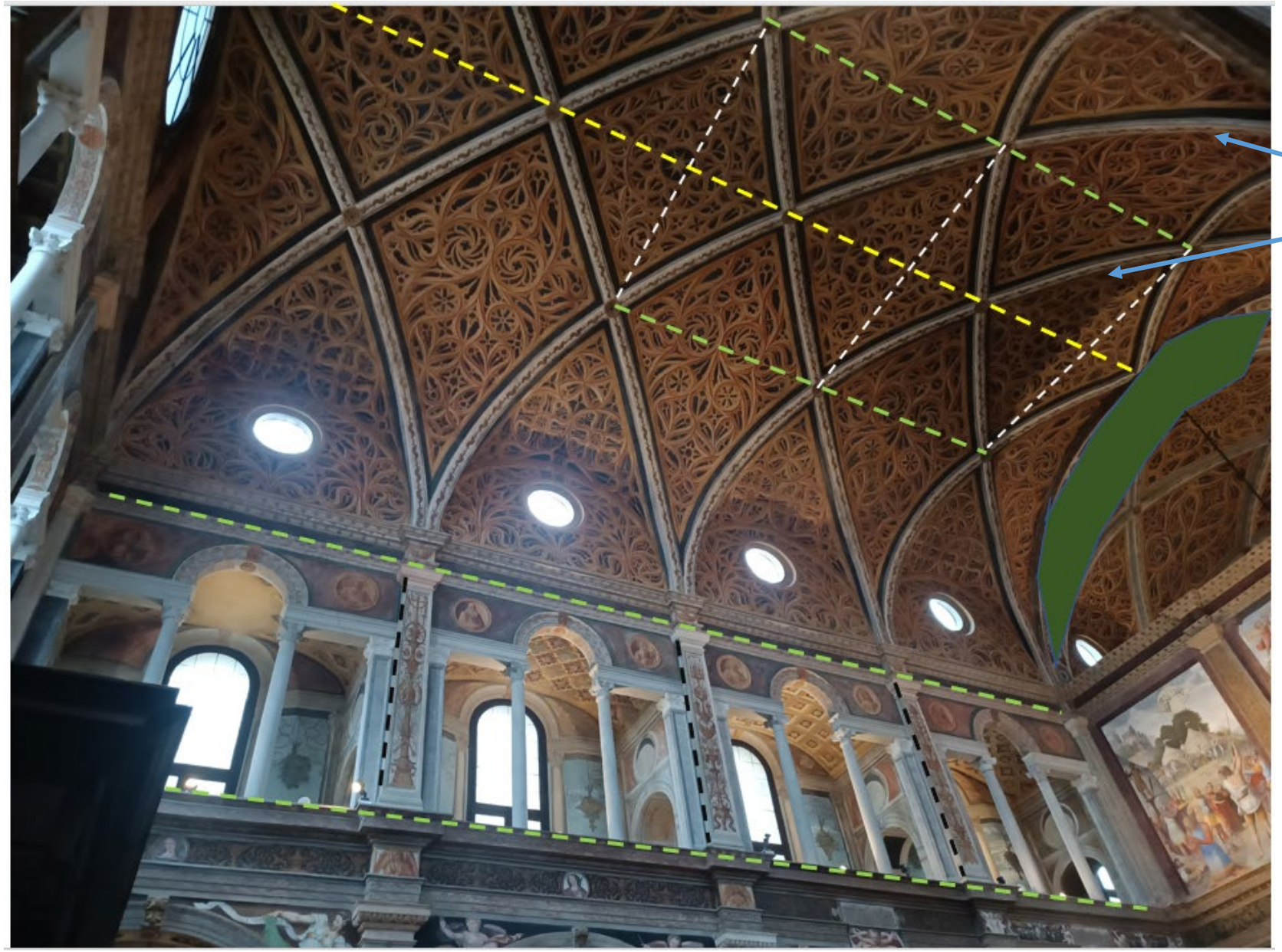
Past homeworks



Palazzo Te - Mantova
cylindric vault



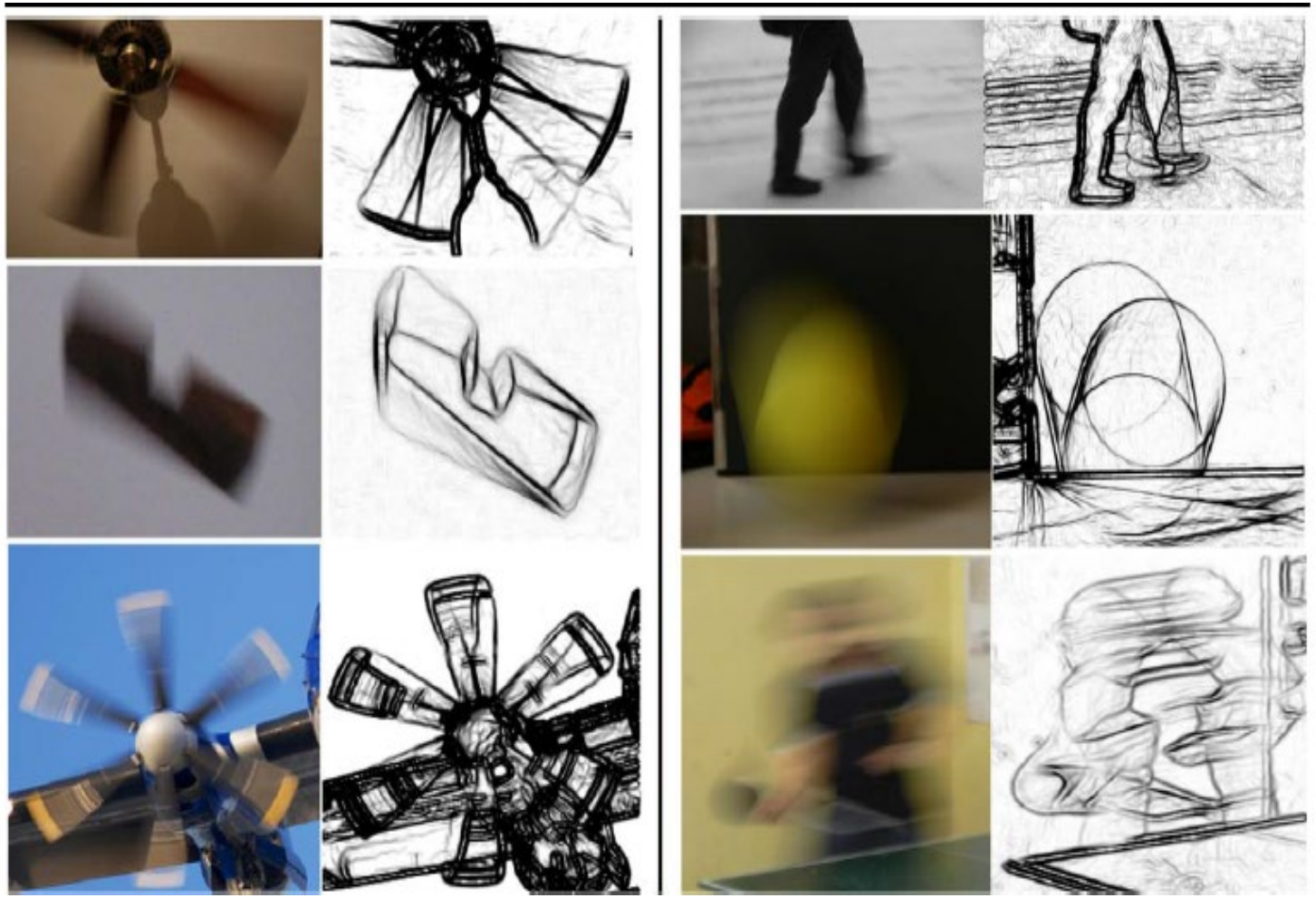




diagonal arcs are parallel but they are not conics

Some past **project topics**

Analysis of motion-blurred images



Visual analysis of moving vehicles

Find USEFUL symmetric points from light contours
1. Bitangents 2. Points colinear to vanishing point

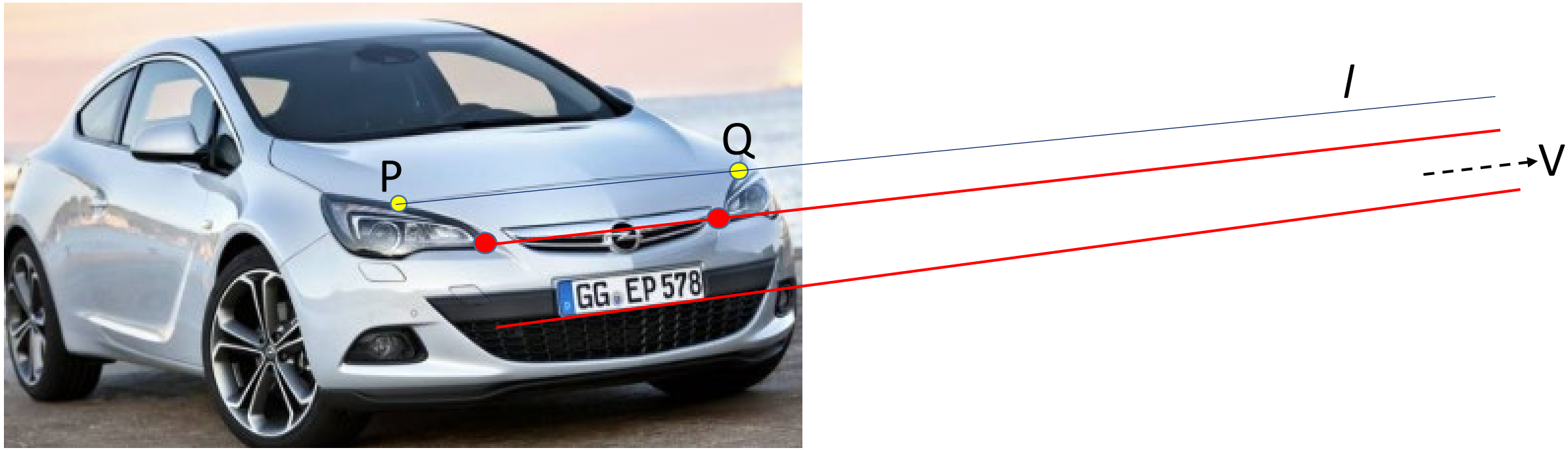


Image of midpoints

Symmetry plane in 3D

Backproject other midpoints onto π

Solve triangle OAB → A B in 3D → C = (A+B)/2

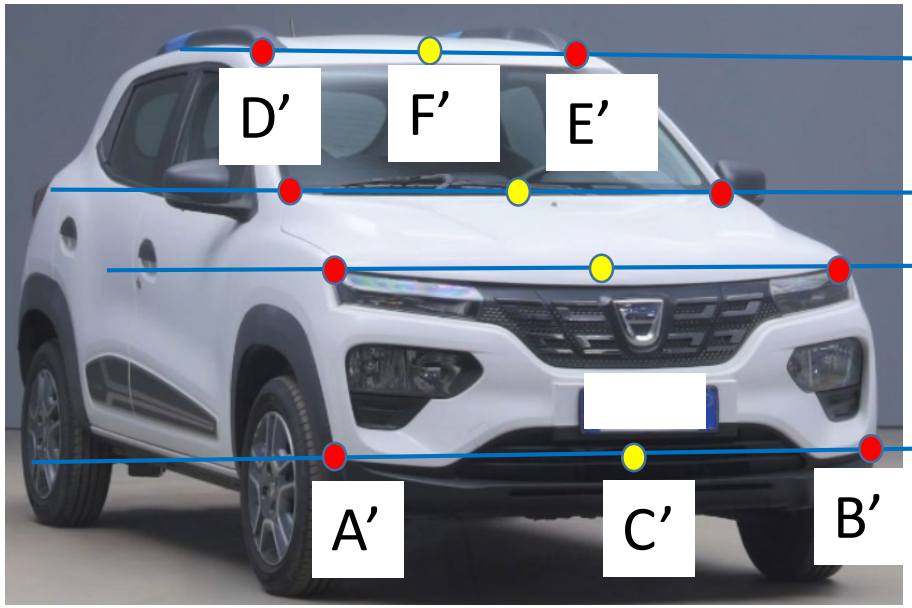
$\boldsymbol{n} = \boldsymbol{K^{-1}V}$ $\quad d = -[X_C \quad Y_C \quad Z_C]\boldsymbol{n}$ $\quad \pi = [\boldsymbol{n}^T \quad d]^T$

$w = -\boldsymbol{n}^T\boldsymbol{K^{-1}F'}/d$ → cartesian coord.s of F $= -\boldsymbol{K^{-1}F'}/w$

3D position of symmetric features

Find 3D positions of symmetric features, e.g. D (or E)

Solve triangle OFD using two known vertices O=(0, 0, 0) and F and known directions OD $= -\boldsymbol{K^{-1}D'}$ and FD $= -\boldsymbol{K^{-1}V}$



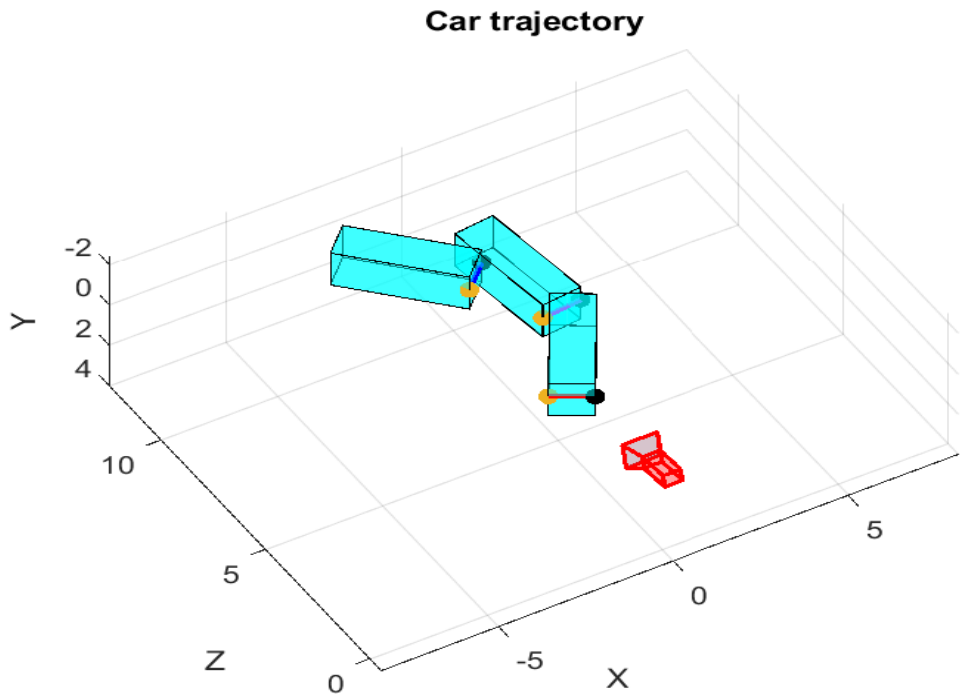
V

symmetric features: reconstructed!!

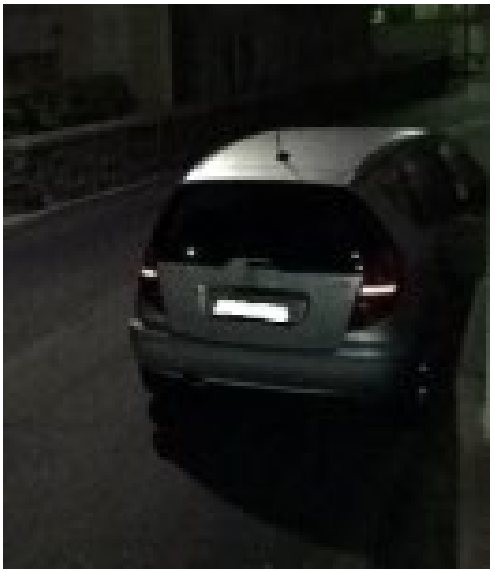

1


2

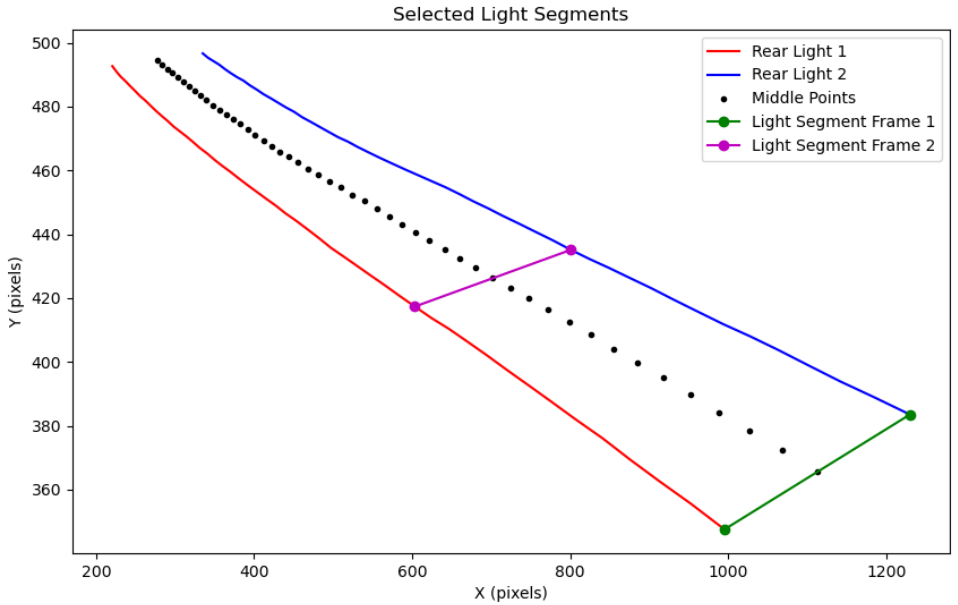

3


Car trajectory
Y
-2
0
2
4
10
5
Z
0
-5
0
5
X

Analysis of vehicle motion in the dark using car lights



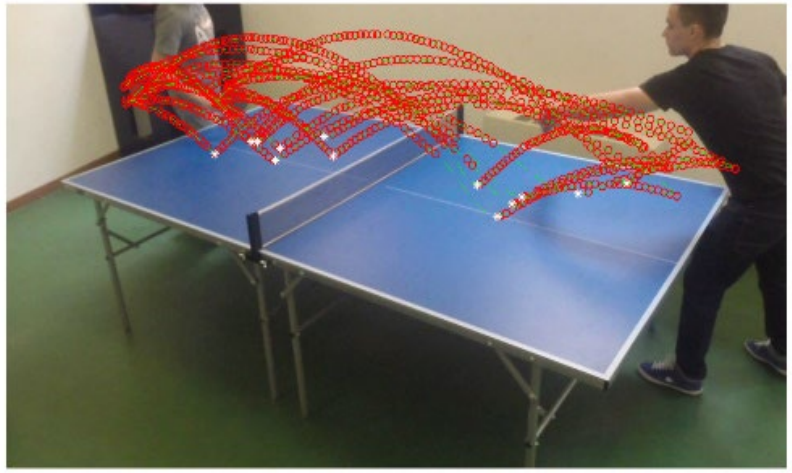


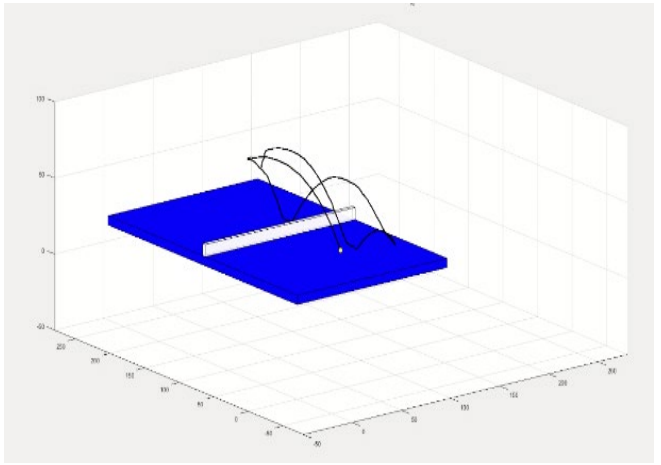


Visual analysis of sport events

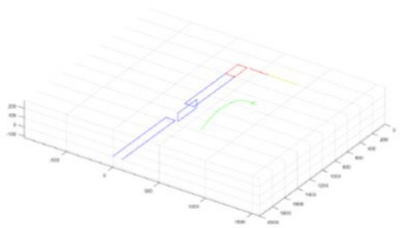


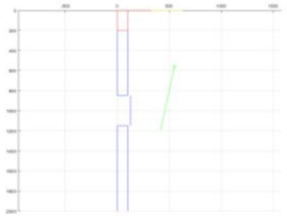




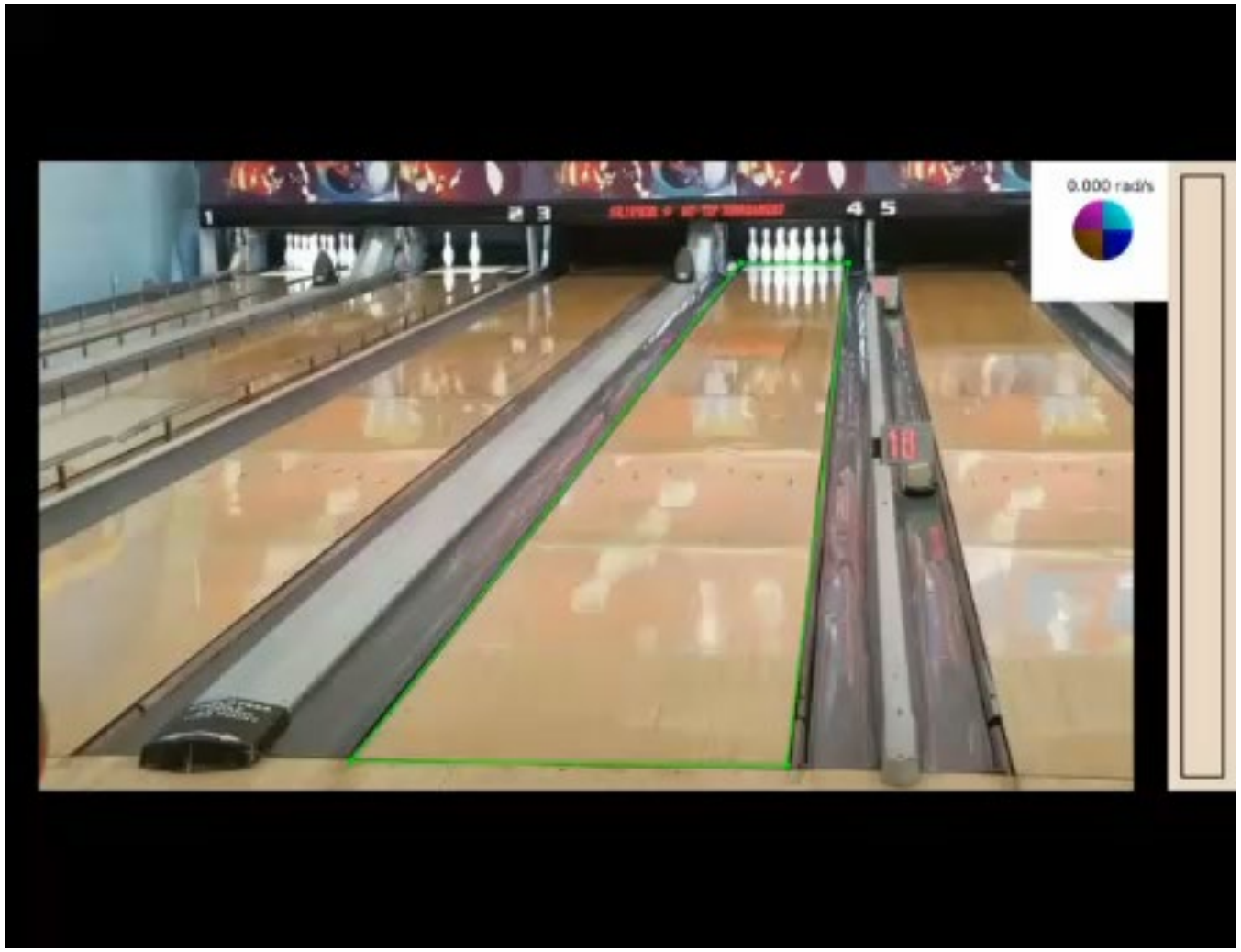




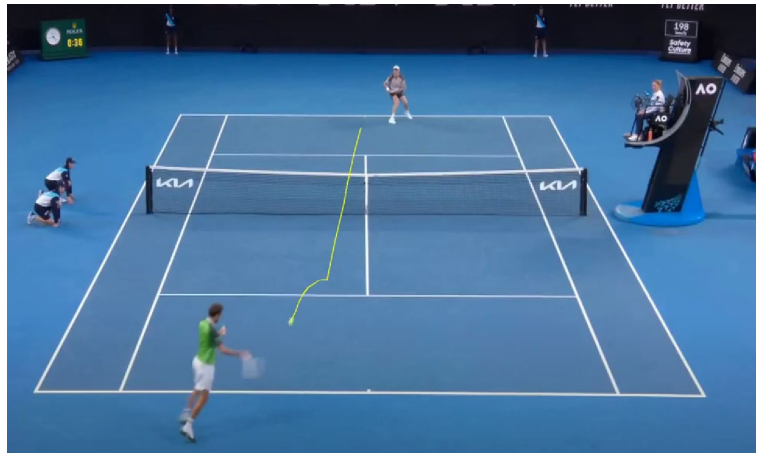


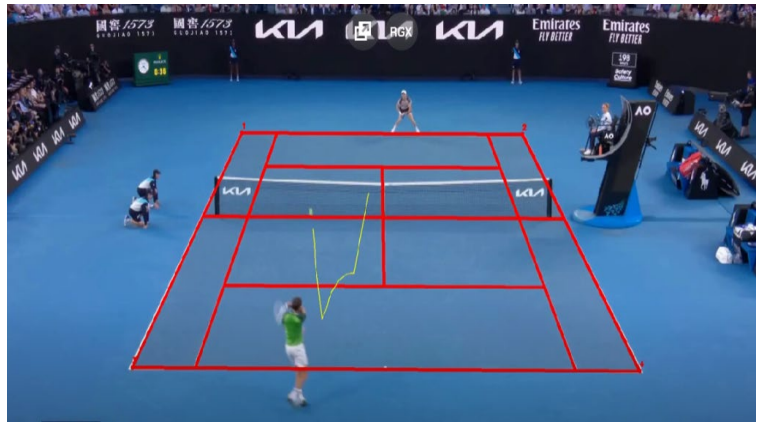


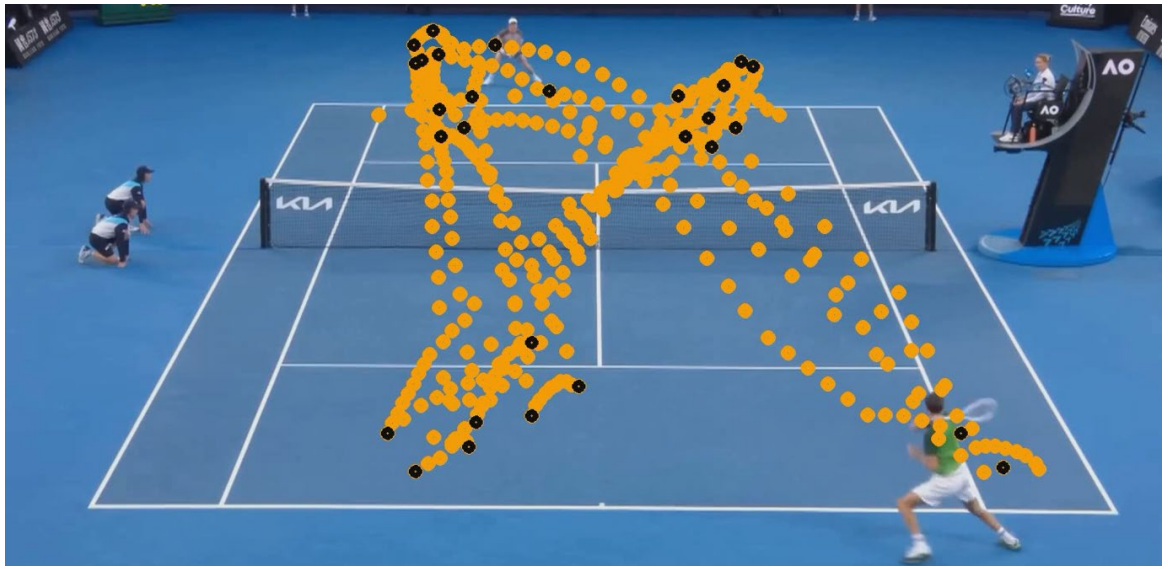
Bowling: ball trajectory and spin

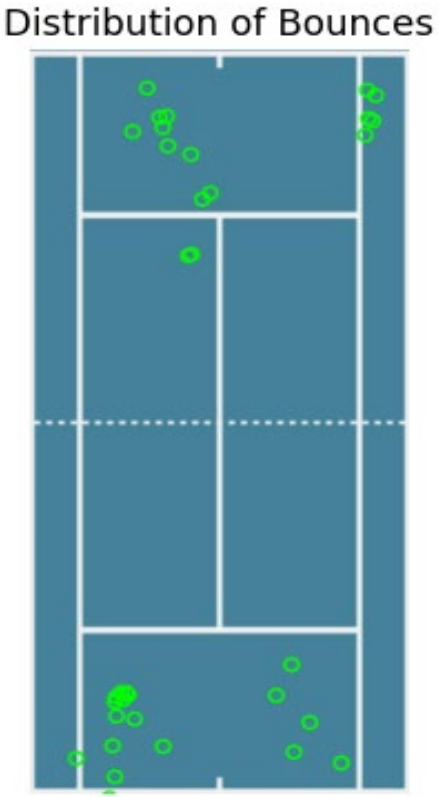


3D trajectory of tennis ball









HOMEWORK

- **HOMEWORK** on Theory:

- A LONG AND DIFFICULT EXERCISE
- INDIVIDUAL (no collaboration)
- MANDATORY IN THE STANDARD EXAM
- PUBLISHED in NOVEMBER
- UNIQUE DEADLINE: within the 1° EXAM CALL (about end of January)
- but still VALID IN FUTURE CALLS AND FUTURE YEARS
- DEVELOPED IN MATLAB
- EVALUATION based on (i) SUBMISSION and (ii) REPORT
- FOLLOW DELIVERY INSTRUCTION: PUBLISHED at the same time as HOMEWORK

PROJECT

- application **PROJECT**:

- TEAMS of 1, 2, or 3 STUDENTS (collaboration is welcome)
- MANDATORY IN STANDARD EXAM
- LIST OF TOPICS: PUBLISHED MID DECEMBER,
- DEVELOPED IN (ALMOST) ANY PROGRAMMING LANGUAGE
- MANY DEADLINES = THREE DAYS BEFORE EACH EXAM CALL
- EVALUATION based on
- (i) SUBMITTED MATERIAL
- (ii) pdf REPORT and ppt PRESENTATION
- (iii) PRESENTATION/DISCUSSION
- FOLLOW DELIVERY INSTRUCTIONS: published at the same time as TOPICS

Project evaluation:
maximum score (20/30) is NOT the DEFAULT outcome.
Here are some minimal requirements:

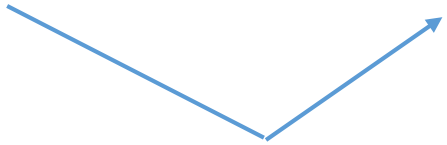
- Awerness of the theory taught in the course: avoid unnecessary assumption or restrictions
- Awereness of the theory taught in the course: avoid to re-invent (often worst) solutions
- Awereness of the theory taught in the course: capability to autonomously apply theory to derive solutions
- Autonomy in the retrieval and/or construction of the needed data sets
- Experimental results based on real images, not only simulated ones
- Development of a solution with a certain degree of generality, without tailoring it to just a single case (e.g., single video or image)

STANDARD EXAM: EVALUATION → 30 cum laude

CHOOSE BETWEEN TWO POSSIBILITIES :

1° HOMEWORK* 10/30 + FULL PROJECT 20/30

2° HOMEWORK* good evaluation (> 8/10) Pass /Fail + FULL PROJECT 30/30



30 cum laude: ONLY IN CASE OF 10/10 homework evaluation and **excellent** project

* the evaluation of the homework is saved for future calls: you don't have to deliver it twice

NON-STANDARD EXAM: MAX GRADE < 30 can be chosen individually, but ONLY upon TIMELY REQUEST

NON-STANDARD EXAM: MAX GRADE < 30

PROJECT ONLY (NO HOMEWORK) → max **28**/30

FULL PROJECT** **20**/30 + WRITTEN PROOF*** (including difficult exercises) **8**/30

HOMEWORK ONLY (NO PROJECT) → max **27**/30

HOMEWORK* **10**/30 + EXT. WRITTEN PROOF*** (including difficult exercises) **17**/30

NO HOMEWORK & NO PROJECT → max total **25**/30

EXTRA-EXT. WRITTEN PROOF*** (including difficult exercises) **25**/30

HOMEWORK & **SHORT** PROJECT** (in place of FULL project) → max total **26**/30

HOMEWORK **10**/30 + SHORT PROJECT **16**/30

* The evaluation of the homework is saved for future calls: you don't have to deliver it twice

** The evaluation of the project is saved for future calls: you don't have to deliver it twice

*** The evaluation of the written proof is NOT valid anymore after the call: you will have to repeat it

NON-STANDARD EXAM:
ONLY ON SPECIFIC REQUEST

within **3 days before** the call date, please:

send an e-mail to vincenzo.caglioti@polimi.it including **request** for nonstandard exam (written proof), where you specify the exam mode:

- PROJECT ONLY also indicate both Project Title and Name of Supervisor (within the same deadline you have also to deliver the project, and to e-mail both Project Report and Project Presentation file*)
- HOMEWORK ONLY
- NO PROJECT & NO HOMEWORK

* THIS IS NOT REDUNDANT since you might be part of a project team, including a team-mate, who is going to take the standard exam

→ need to know how many written proofs to prepare

NON-STANDARD EXAM:
WARNING

- I DON’T RECEIVE any timely request of nonstandard exam from you

→ you will not be admitted to participate to that call

- NO SHOW after you sent a request for written proof (nonstand. exam)

→ *RIPROVATO*: a negative mark that will remain in your files and prevents your participation to the subsequent calls in the same exam session

please NOTE:
different team members
can take the exam in different calls